

AC to DC Converter (Design-01)

Electronics Design Notes

1 Overview

This document explains the design of an **AC to DC converter circuit** and the corresponding **PCB design and routing guidelines**. The system converts an AC input into a usable DC output using rectification, filtering, and indication stages.

2 AC to DC Converter Circuit Explanation

2.1 Input Stage

- The input (IP) is an AC supply.
- The AC input is connected to a bridge rectifier.

2.2 Bridge Rectifier

- Converts AC voltage into DC voltage.
- The output obtained is pulsating DC.
- Pulsating DC is not suitable for direct use in electronic circuits.

2.3 Filtering Stage

- A capacitor is connected after the bridge rectifier.
- The capacitor smoothens the pulsating DC.
- This converts pulsating DC into near-pure DC.
- A THT electrolytic capacitor is generally used.

2.4 Discharge Resistor

- A resistor is connected across the capacitor.
- It discharges the capacitor when the supply is switched OFF.
- Improves circuit safety and reliability.

2.5 LED Indicator

- An LED is used for power ON indication.
- A series resistor limits current through the LED.

3 Design Rule Check (DRC)

- DRC stands for Design Rule Check.
- It checks for electrical and manufacturing rule violations.
- Helps detect missing connections, shorts, and clearance errors.

4 PCB Footprints

- Correct footprints must be assigned to all components.
- Electrolytic capacitors use Through-Hole Technology (THT).
- A $0\ \Omega$ resistor is used as a jumper.
- It helps in horizontal THT routing and layout flexibility.

5 PCB Editor Guidelines

5.1 Edge Cuts

- The Edge Cut layer is used to draw the PCB outline.
- It defines the physical dimensions of the PCB.

6 Routing Phase

6.1 Track Width Selection

- Track width must be chosen based on current requirements.
- Thicker tracks are required for higher current.
- Path to change track width:
 - File → Board Setup → Design Rules → Predefined Sizes → Width
- Commonly used widths:

- 0.5 mm for low-current signals
- 1 mm for AC or high-current lines

6.2 Layer Configuration

- A 6-layer PCB configuration is selected.
- AC signal traces should preferably use thicker track widths.

6.3 Copper Removal Tool

- Removes copper only from selected areas.
- Prevents complete copper removal from the PCB.
- Assists the fabricator during PCB manufacturing.

7 Adding Logos on PCB

- Logos or text can be added using the Add Footprint option.
- This option is available in the PCB Editor side bar.

8 Conclusion

The AC to DC converter design demonstrates the complete process from rectification to PCB layout. Proper component selection, routing rules, and design checks ensure a reliable and manufacturable PCB.